

PROPERTIES OF SOLUTIONS

Chapter 13

13.1 Solution Process

13.2 Saturated solution and solubility

13.3 Factors Affecting Solubility

13.4 Expressing Solution Concentration

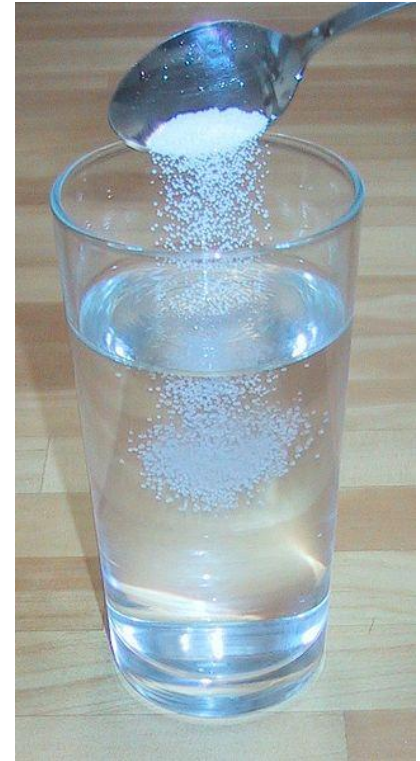
13.5 Colligative Properties



Solutions

- A **solution** is a homogeneous mixture of two or more substances.
- The substance present in the greatest quantity is usually called the **solvent** and the other substances are called **solutes**.

Solute + Solvent → Solution



Heterogeneous mixture

Homogeneous solution



**Completely
Ionized**



**Conduct
electricity**

**Strong acids
Ionic compounds**

**Strong
Electrolyte**

**Partially
Ionized**



**Conduct
electricity**

**Weak acids
Weak bases**

**Weak
Electrolyte**

**Does not
Ionized**



**Does not conduct
electricity**

**All other
compounds**

**Non
Electrolyte**



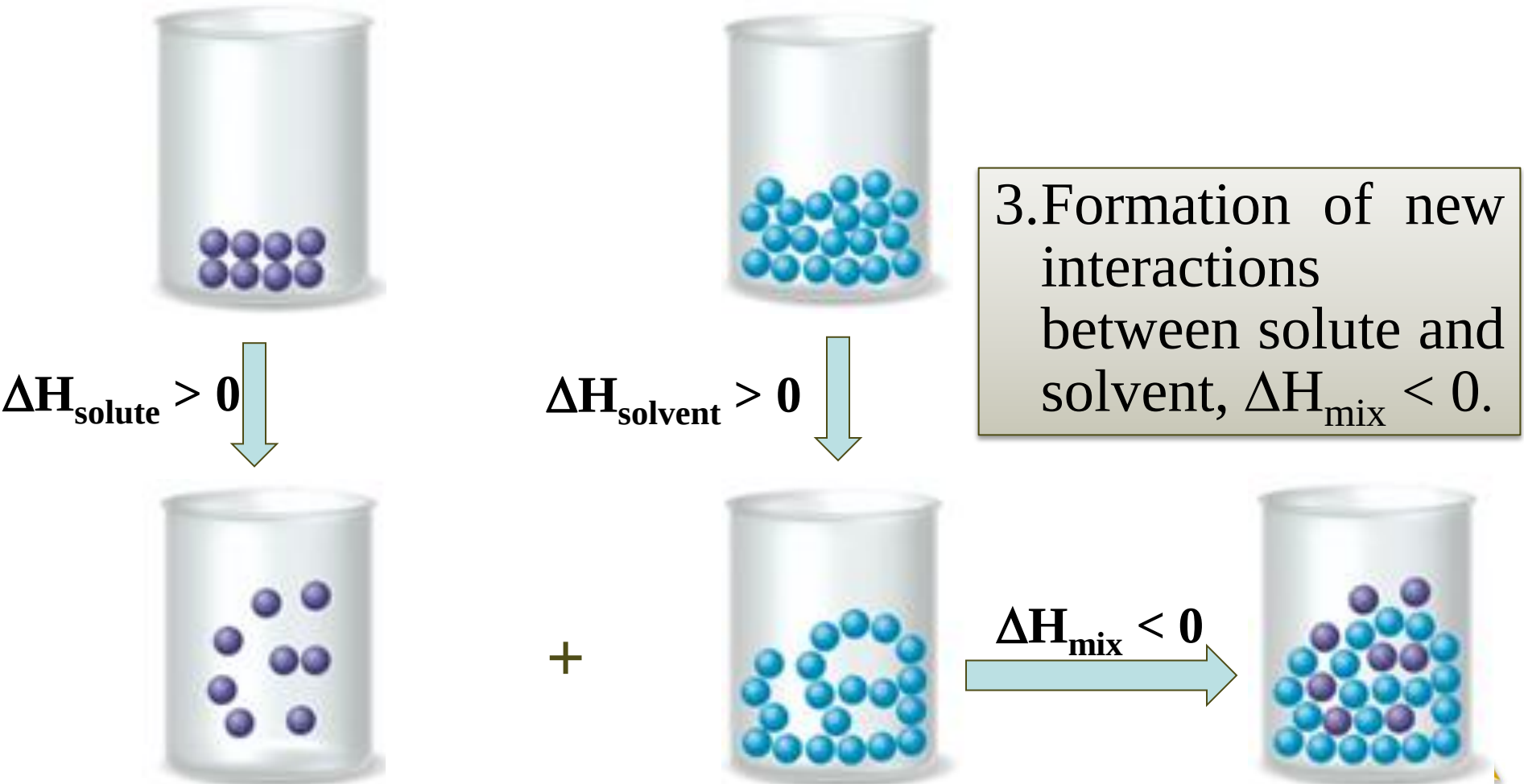
13.1 Solution Process

The solution process involves three steps:

1. Separation of solute particles, $\Delta H_{\text{solute}} > 0$.

2. Separation of solvent particles to make 'holes', $\Delta H_{\text{solvent}} > 0$.

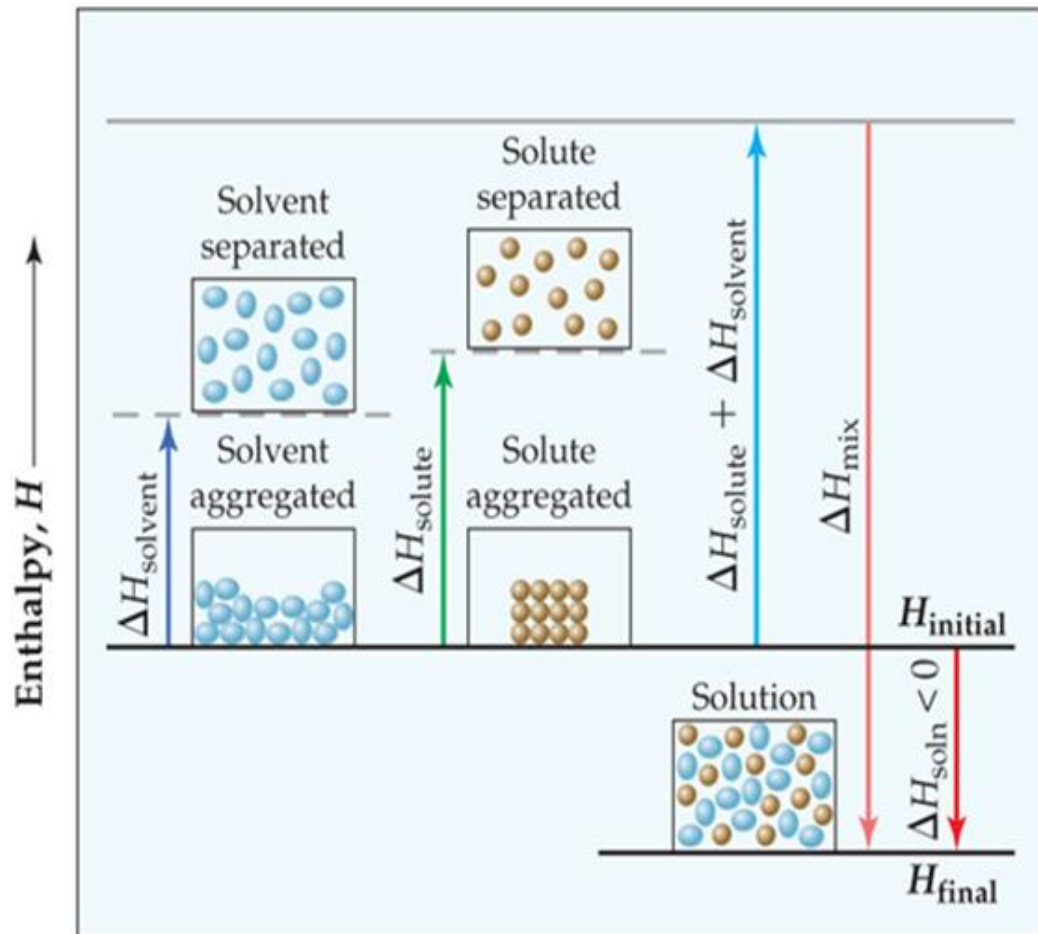
3. Formation of new interactions between solute and solvent, $\Delta H_{\text{mix}} < 0$.



$$\Delta H_{\text{soln}} = \Delta H_{\text{solvent}} + \Delta H_{\text{solute}} + \Delta H_{\text{mix}}$$

- If $\{\Delta H_{\text{solvent}} + \Delta H_{\text{solute}}\} < \Delta H_{\text{mix}}$, then $\Delta H_{\text{soln}} < 0$

- The solution process is enthalpy favorable, $\text{MgSO}_4(\text{aq})$, -91.2 kJ/mol .
- The magnitude of solvent-solute interaction are either comparable or greater than the solute-solute and solvent-solvent interactions.

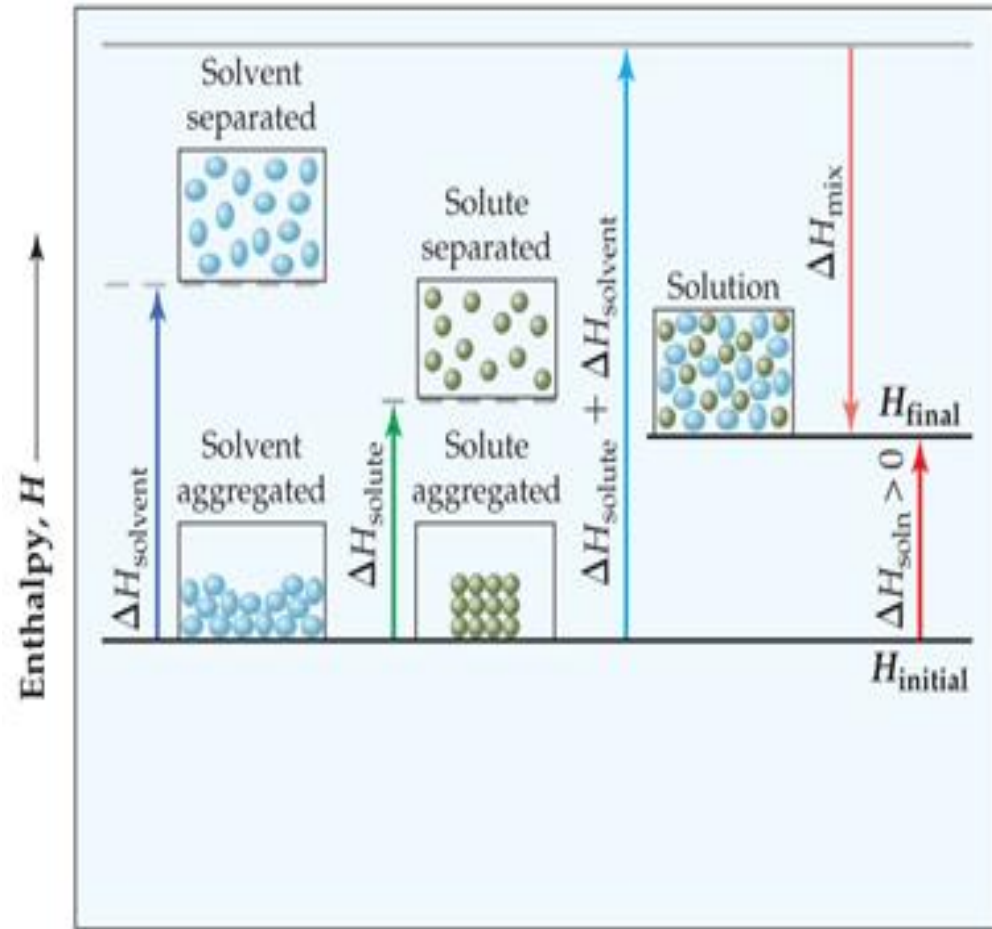


$$\Delta H_{\text{soln}} = \Delta H_{\text{solvent}} + \Delta H_{\text{solute}} + \Delta H_{\text{mix}}$$

- If $\{\Delta H_{\text{solvent}} + \Delta H_{\text{solute}}\} > \Delta H_{\text{mix}}$, then $\Delta H_{\text{soln}} > 0$

➤ The solution process is enthalpy un-favorable, $\text{NH}_4\text{NO}_3(\text{aq})$, 26.4 kJ/mol.

➤ The magnitude of solvent-solute interactions are weaker than the solute-solute and solvent-solvent interactions.

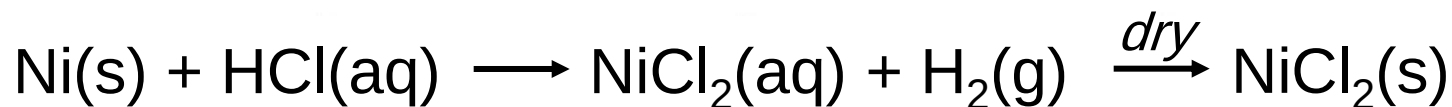


- **Why do endothermic processes sometimes occur spontaneously?**

- Some processes, like the dissolution of NH_4NO_3 in water, are spontaneous at room temperature even though heat is absorbed, not released.
- Enthalpy is only part of the picture, the other part is the **ENTROPY** change.
- The formation of solutions is favored by the increase in entropy that accompanies mixing.



Solution formation verses chemical reaction



- Solution is a physical change you can get back the original solute by evaporating the solvent.
- If you can't, the substance didn't dissolve, it reacted.



13.2 Saturated solution and solubility

1. **Saturated Solution:** a solution that contains the maximum amount of solute that can be dissolved into the solvent at a given temperature.
 - **Solubility:** the maximum amount of the solute that can dissolve in a given amount of the solvent at a specified temperature.
2. **Supersaturated Solution:** a solution can contain more solute than needed to form a saturated solution by raising the temperature of the solvent.
 - These solutions are unstable; crystallization can usually be stimulated by adding a “seed crystal” or scratching the side of the flask.
 - Crystallization is the opposite of the solution process

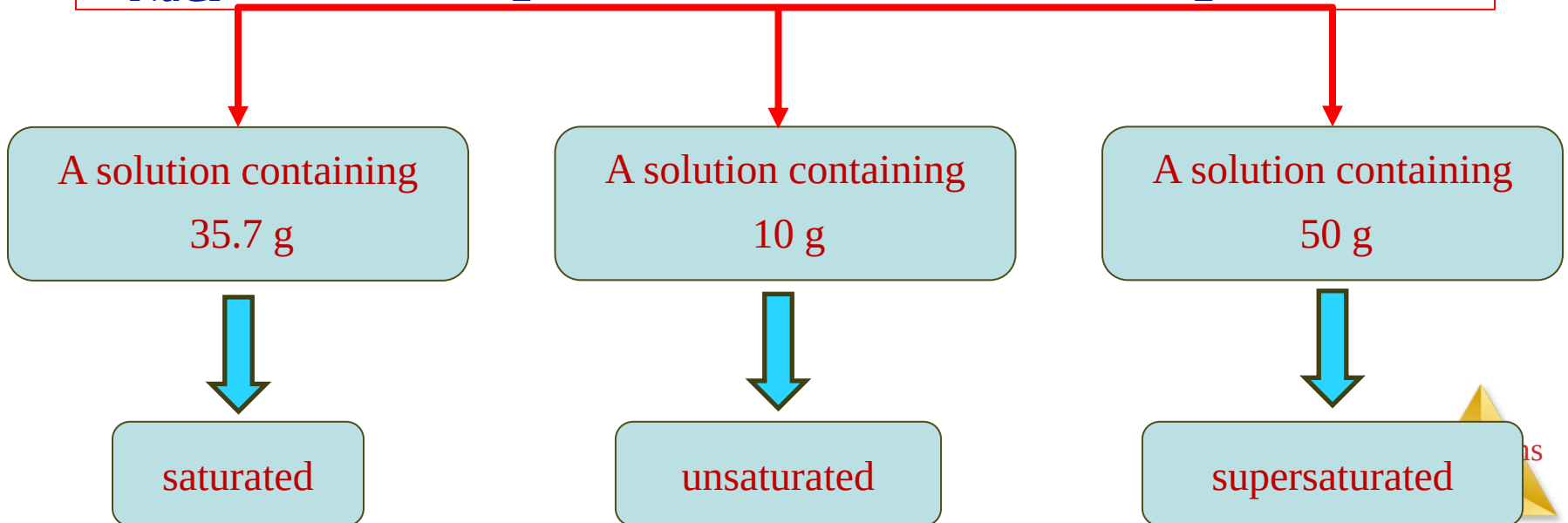




- A solution that is in equilibrium with undissolved solute is saturated.

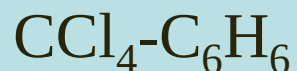
3. **Unsaturated Solution:** contains less than the maximum amount of solute that can dissolve in a given amount of the solvent at a specified temperature.

S_{NaCl} in water H_2O = 35.7 g per 100 mL H_2O at 0°C .



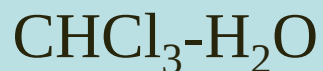
Intermolecular Interactions

Dispersion



between all molecules, whether they are polar or nonpolar

Dipole-dipole



between the positive end of one polar molecule and the negative end of another polar molecule

Hydrogen bond



between a H atom which is bound to a more electronegative atom or group, such as N, O, or F

Ion-dipole



between an ion and a neutral molecule that has a dipole.

Solutions

polar liquids tend to dissolve in
polar solvents

nonpolar liquids tend to dissolve
in nonpolar solvents

like dissolves like

water
H₂O



ammonia
NH₃



Soluble
(miscible)

**Polar and able to
form H-bonds**

**Polar and able to
form H-bonds**

- **Miscible: pairs of liquids that mix in all proportions.**

water
 H_2O



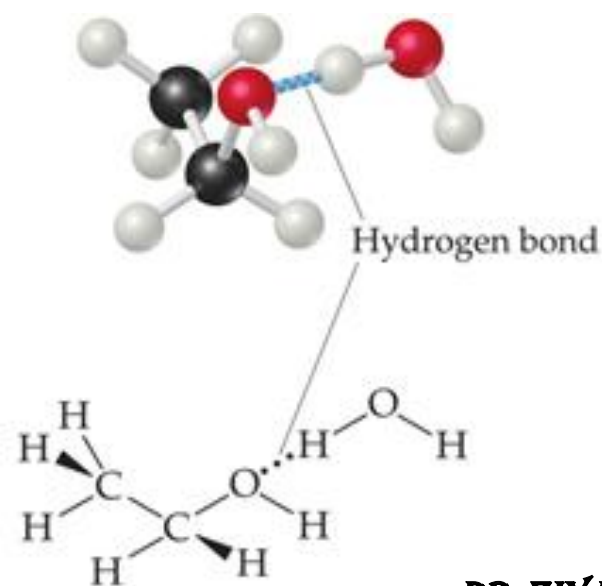
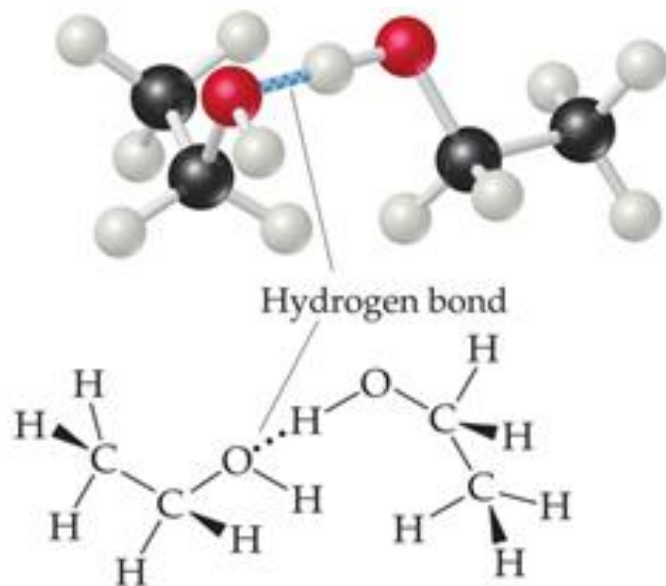
Acetone
 CH_3COCH_3



Soluble
(miscible)

Polar and able to form H-bonds Has polar $\text{C}=\text{O}$ and pairs of non-bonding electrons in O atom that can form H-bonds with water

Example: ethanol in water



Solutions

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water
 H_2O



Hexane
 C_6H_{14}



Insoluble
(immiscible)

polar

nonpolar

- immiscible: pairs of liquids that do not dissolve in one another.

Nonpolar liquids tend to be insoluble in polar liquids.

I_2



CS_2



soluble

nonpolar

nonpolar

Br_2



C_6H_6



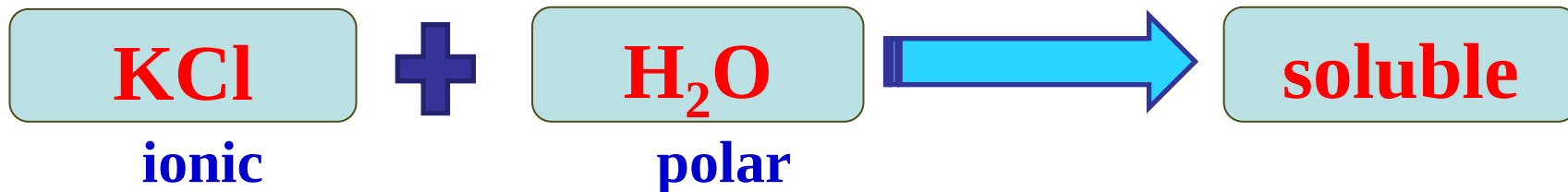
soluble

nonpolar

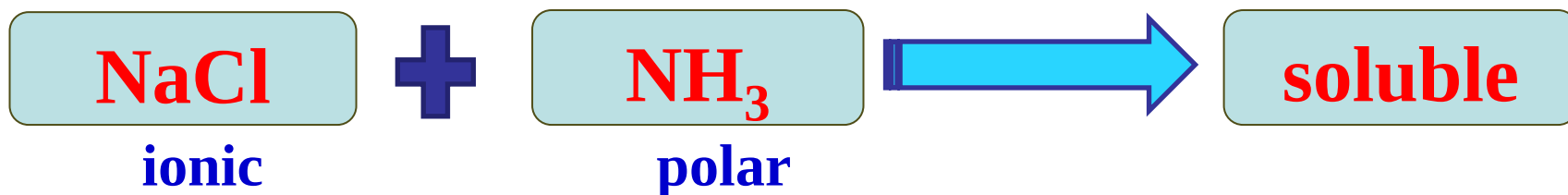
nonpolar

Solutions

DR. ZIYAD A. TAHA



- Hydration: the process in which an ion or a molecule is surrounded by water molecules.



- Solvation: the process in which an ion or a molecule is surrounded by solvent molecules.

Br₂ in C₆H₆	more soluble	Br₂ in H₂O
KCl in CCl₄	less soluble	KCl in NH₃
CH₂O in CS₂	less soluble	CH₂O in H₂O
I₂ in CS₂	more soluble	I₂ in H₂O



13.3 Factors Affecting Solubility

1. Solute-Solvent Interactions

- In general, when other factors are comparable, the stronger the attractions between solute and solvent molecules, the greater the solubility of the solute in the solvent.
- In general, the solubility of gases in water increases with increasing mass, Why?
- Larger molecules have stronger dispersion forces.

Gas	Solubility, M	M. mass
N ₂	0.69 x 10 ⁻³	28
CO	1.04 x 10 ⁻³	28
O ₂	1.38 x 10 ⁻³	32
Ar	1.50 x 10 ⁻³	39.9
Kr	2.79 x 10 ⁻³	83.8

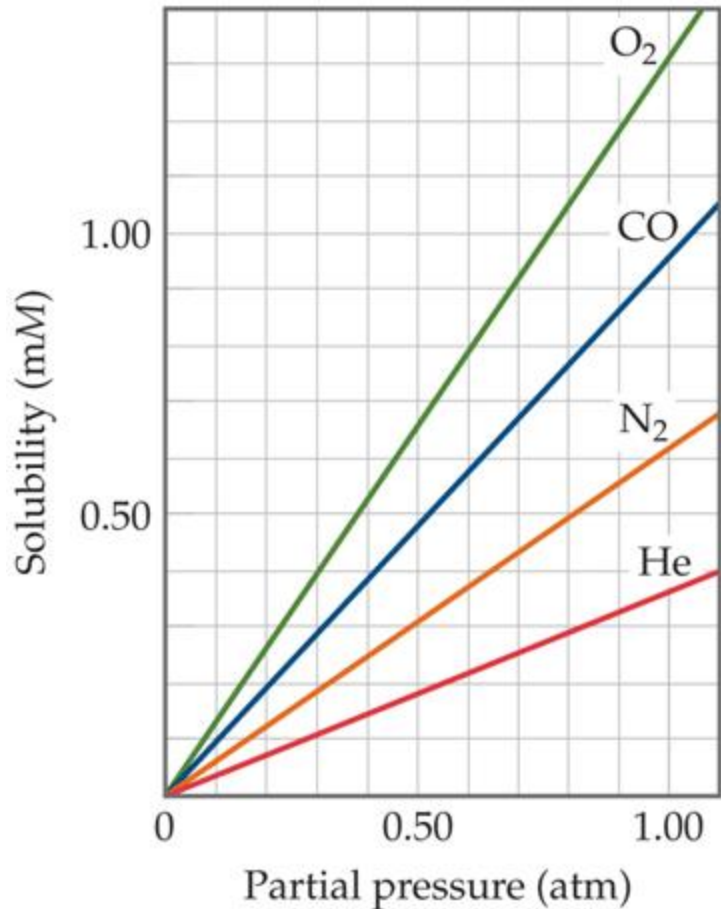
TABLE 13.3 Solubilities of Some Alcohols in Water and in Hexane*

Alcohol	Solubility in H ₂ O	Solubility in C ₆ H ₁₄
CH ₃ OH (methanol)	∞	0.12
CH ₃ CH ₂ OH (ethanol)	∞	∞
CH ₃ CH ₂ CH ₂ OH (propanol)	∞	∞
CH ₃ CH ₂ CH ₂ CH ₂ OH (butanol)	0.11	∞
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ OH (pentanol)	0.030	∞
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ OH (hexanol)	0.0058	∞
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ OH (heptanol)	0.0008	∞

*Expressed in mol alcohol/100 g solvent at 20°C. The infinity symbol indicates that the alcohol is completely miscible with the solvent.

- As the number of C atoms in an alcohol increases, the polar OH group becomes an ever smaller part of the molecule, and the molecule behaves more like a hydrocarbon {the solubility in water decreases with increasing number of C atoms}.
- The solubility of alcohols in a nonpolar solvent like hexane increase's as the nonpolar hydrocarbon chain lengthens.

Pressure Effects



- The solubility of liquids and solids does not change appreciably with pressure.
- The solubility of a gas in a any solvent is increased as the partial pressure of the gas above the solvent increases.

solubility
of the gas,
M

$$S_g = kP_g$$

Henry's Law

Henry's Law
constant, M/atm

the partial
pressure of the
gas over the
solution, atm.

- The value of K depends on the solute, solvent and temperature.

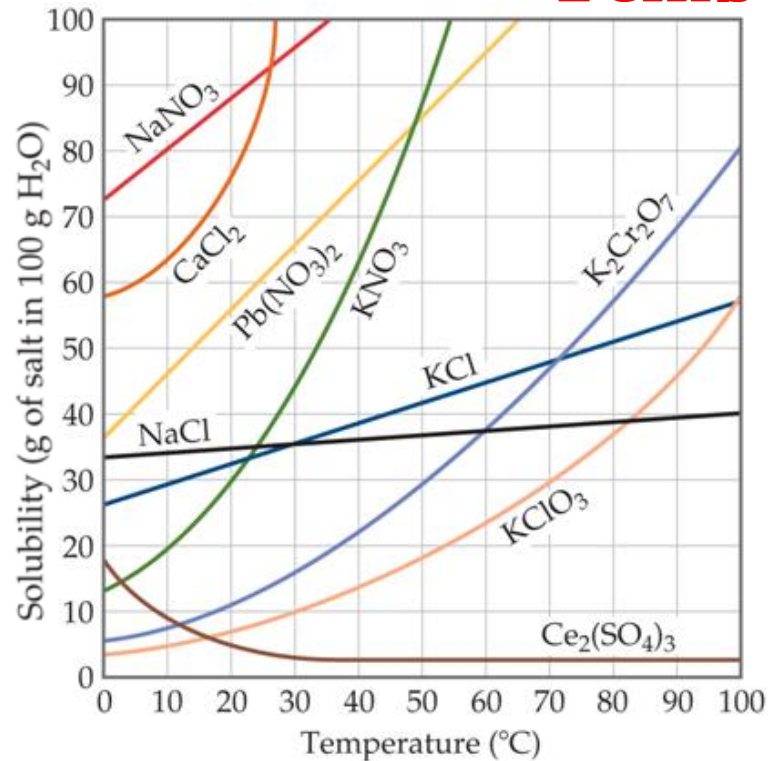
Sample Exercise

A certain soft drink is bottled so that a bottle at 25°C contains CO₂ gas at a pressure of 5.0 atm over the liquid. Assuming that the partial pressure of CO₂ in the atmosphere is 4.0×10^{-4} atm, calculate the equilibrium concentrations of CO₂ in the soda both before and after the bottle is opened. The Henry's law constant for CO₂ in aqueous solution is 0.031 M/atm at 25°C.

before = 0.16 M

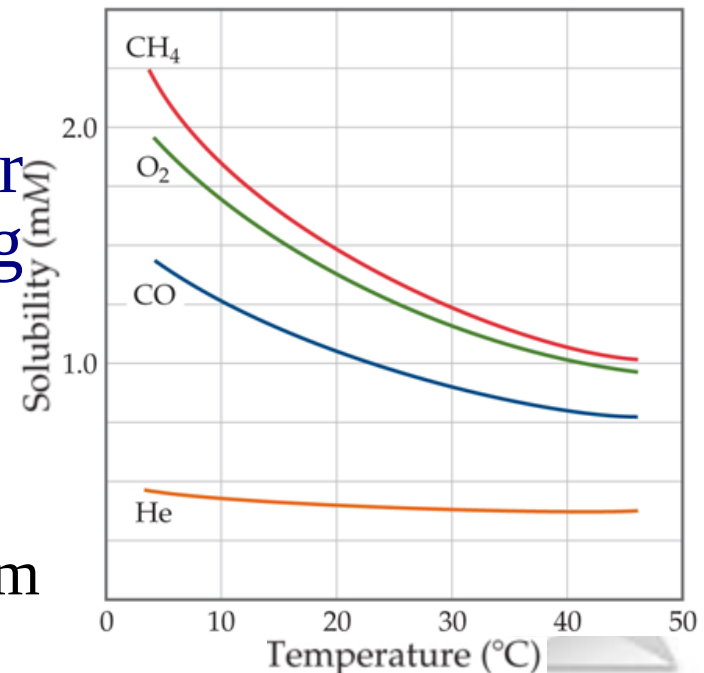
after = 1.2×10^{-5} M

Temperature Effects



- Generally, the solubility of solid solutes in liquid solvents increases with increasing temperature.
- Solubility of Ce₂(SO₄)₃ decreases with temperature.
- Most solution formation is endothermic.

- The solubility of gases in water decreases with increasing temperature.
 - Carbonated soft drinks are more “bubbly” if stored in the refrigerator
 - Warm lakes have less O₂ dissolved in them than cool lakes.

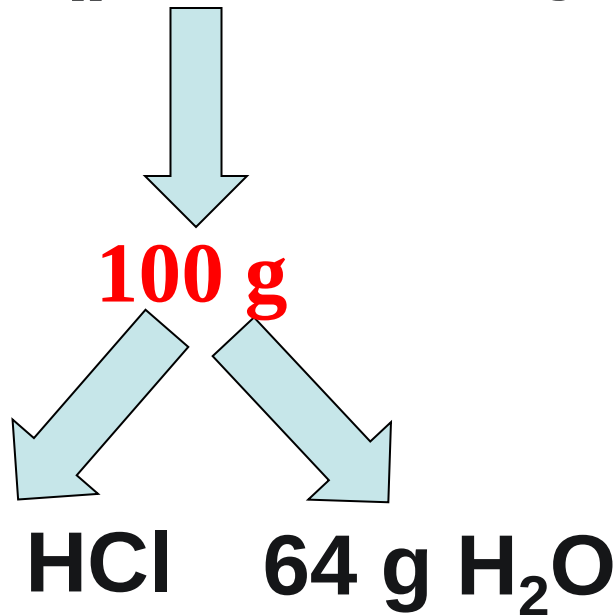


13.4 Expressing Solution Concentration

1. Mass Percent

$$\text{percent by mass} = \frac{\text{mass component}}{\text{mass solution}} \times 100\%$$

36% HCl(aq) solution by mass

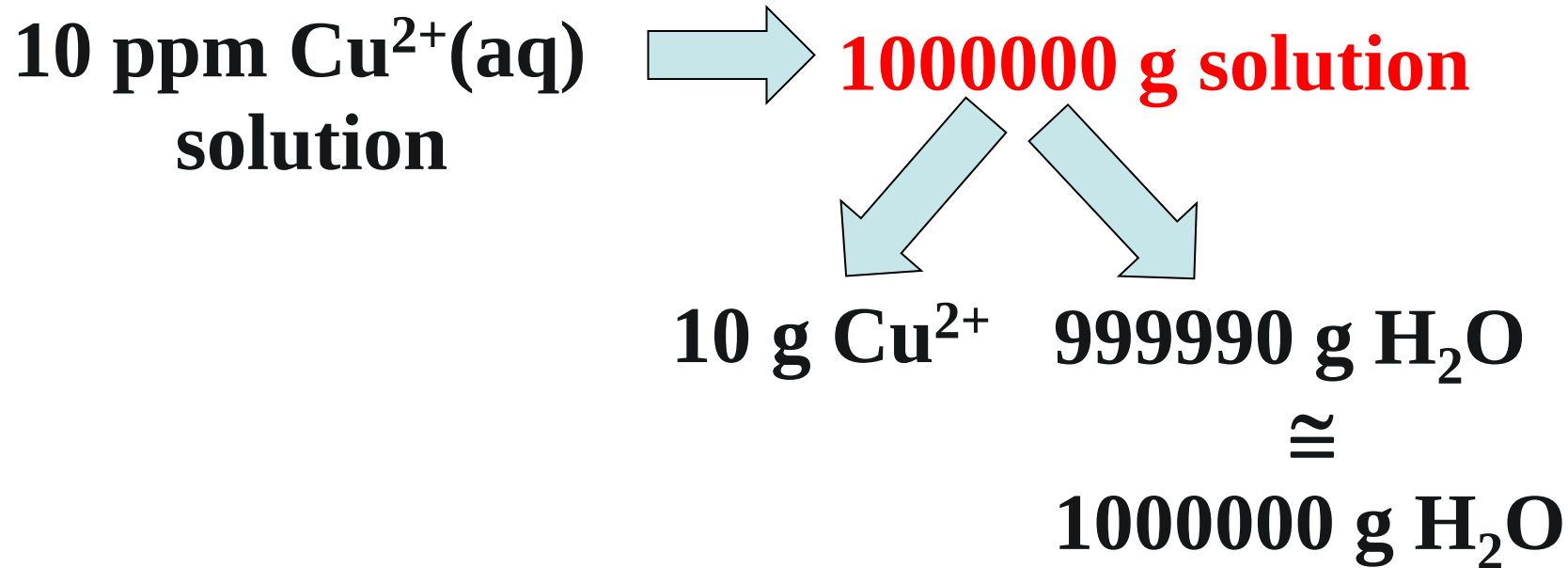


$$\text{mass}_1\% + \text{mass}_2\% + \dots = 100$$



2. Parts Per Million, ppm

$$\text{ppm} = \frac{\text{mass component}}{\text{mass solution}} \times 10^6$$



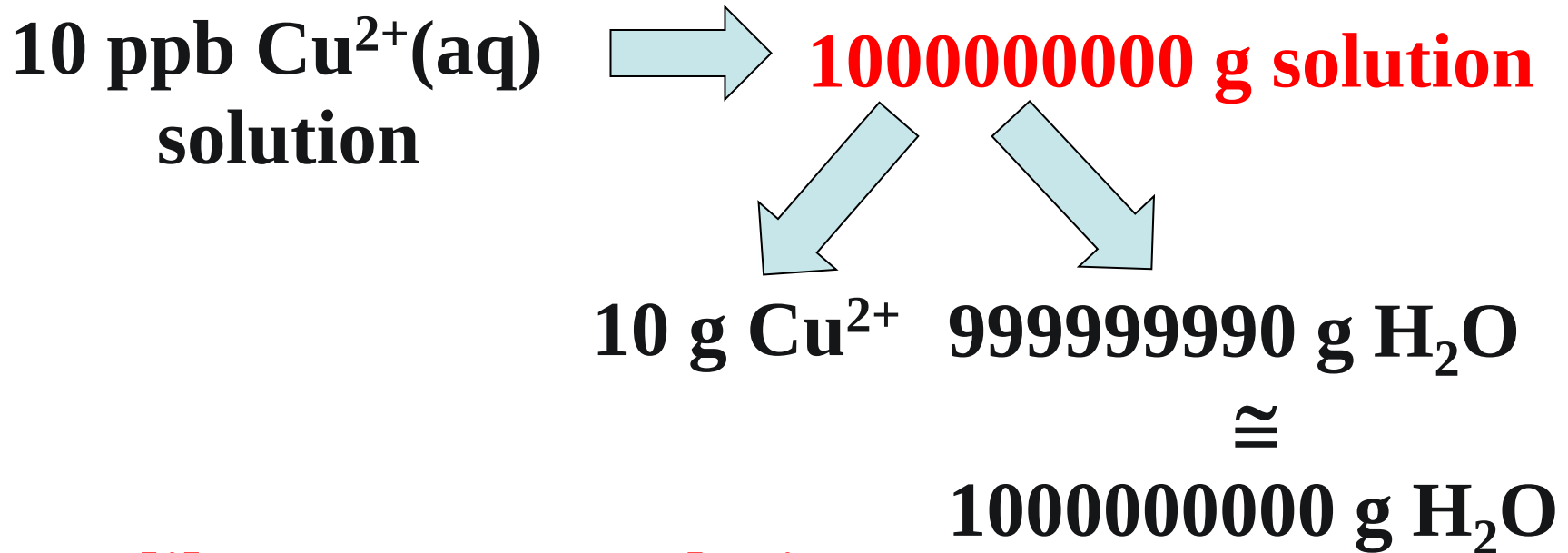
Very dilute aqueous solutions:

$$1 \text{ ppm} = 1 \mu\text{g/g} = 1 \text{ mg/Kg} = 1 \text{ mg/L} = 1 \mu\text{g/mL}$$



3. Parts Per Billion, ppb

$$\text{ppb} = \frac{\text{mass component}}{\text{mass solution}} \times 10^9$$



Very dilute aqueous solutions:

$$1 \text{ ppb} = 1 \mu\text{g/Kg} = 1 \mu\text{g/L} = 1 \text{ ng/mL} = 1 \text{ ng/g}$$



4. Mole Fraction (χ)

$$\chi = \frac{\text{mol component}}{\text{total mol}} = \frac{n_i}{n_t}$$

$$n_{\text{solute}} = 0.95 \text{ mol}$$

$$\chi_{\text{solute}} = 0.95 \rightarrow \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}} = \frac{0.95}{1.0}$$

$$n_{\text{solute}} + n_{\text{solvent}} = 1.0 \text{ mol}$$

$$n_{\text{solvent}} = 1.0 - 0.95 = 0.05 \text{ mol}$$

$$\chi_{\text{Solute}} + \chi_{\text{solvent}} = 1.0$$



5. Molarity (M)

Molarity: number of moles of solute per liter of solution.

$$M = \frac{\text{moles solute}}{\text{L solution}} \rightarrow \frac{\text{mol}}{\text{L}} \equiv \frac{\text{mmol}}{\text{mL}}$$

$$\text{Number of mol} = M \times V (\text{L})$$

$$\text{Number of mmol} = M \times V (\text{mL})$$



6. Molality (M)

Molality: number of moles of solute per kg of solvent.

$$m = \frac{\text{moles solute}}{\text{kg solvent}} \rightarrow \frac{\text{mol}}{\text{kg}}$$

Number of mol = $m \times \text{kg solvent}$

$$m = \frac{\text{moles solute}}{\text{kg solvent}}$$

$$M = \frac{\text{moles solute}}{\text{L solution}}$$

1.5 mol NaCl

3.0 mol NaCl

3.0 m
NaCl(aq)

0.5 kg H₂O

1.0 kg H₂O



Ways of Expressing Concentrations of Solutions

$$\text{percent by mass} = \frac{\text{mass component}}{\text{mass solution}} \times 100\%$$

$$\text{ppm} = \frac{\text{mass component}}{\text{mass solution}} \times 10^6$$

$$\text{ppb} = \frac{\text{mass component}}{\text{mass solution}} \times 10^9$$

$$\chi = \frac{\text{mol component}}{\text{total mol}} = \frac{n_i}{n_t}$$

$$m = \frac{\text{moles solute}}{\text{kg solvent}}$$

$$M = \frac{\text{moles solute}}{\text{L solution}}$$



**Temperature
dependent**



A 2.5-g sample of groundwater was found to contain $5.4 \mu\text{g}$ of Zn^{2+} . What is the concentration of Zn^{2+} in **ppm**

Answers: 2.2 ppm



What is the **molality** of a solution made by dissolving 36.5 g of naphthalene (C_{10}H_8 , $\mathcal{M} = 128 \text{ g/mol}$) in 425 g of toluene (C_7H_8)?

Answers: 0.670 m



A commercial bleach solution contains 3.62 mass % NaOCl in water. Calculate the **mole fraction** of NaOCl ($\mathcal{M} = 83.5 \text{ g/mol}$).

Answers: 9.00×10^{-3}



A commercial bleach solution contains 3.62 mass % NaOCl in water. Calculate the **molality** of NaOCl ($\mathcal{M} = 83.5 \text{ g/mol}$) in the solution.

Answers: 0.505 m



The concentration of $\text{Pb}(\text{NO}_3)_2$ in a 0.626 M solution is _____
molal. The density of the solution is 1.202 g/mL.

Answers: 0.629



A sulfuric acid solution containing 571.6 g of H_2SO_4 per liter of solution has a density of 1.329 g/mL. Calculate

(a) **the mass percent**

$$\text{mass}\% = \frac{\text{mass solute}}{\text{mass solution}} \times 100\%$$

$$\text{mass salt} = 571.6 \text{ g}$$

$$\text{mass soln} = 1000. \text{ mL} \times 1.329 \text{ g/mL} = 1329 \text{ g}$$

$$\text{mass}\% = \frac{571.6 \text{ g}}{1329 \text{ g}} \times 100\% = \boxed{43.01\%}$$



(b) the mole fraction

$$X_{\text{H}_2\text{SO}_4} = \frac{\text{mol}_{\text{H}_2\text{SO}_4}}{\text{mol}_{\text{H}_2\text{SO}_4} + \text{mol}_{\text{H}_2\text{O}}} = \frac{5.833}{5.833 + 42.03} = 0.1219$$

$$n_{\text{H}_2\text{SO}_4} = \frac{571.6 \text{ g}}{98.0 \text{ g/mol}} = 5.833 \text{ mol}$$

mass H_2O = mass of soln - mass of solt

$$\text{mass } \text{H}_2\text{O} = 1329 - 571.6 = 757.4 \text{ g } \text{H}_2\text{O}$$

$$n_{\text{H}_2\text{O}} = \frac{757.4 \text{ g}}{18.02 \text{ g/mol}} = 42.03 \text{ mol}$$



(c) the molality

$$m = \frac{\text{moles solute}}{\text{kg solvent}} = \frac{5.833 \text{ mol H}_2\text{SO}_4}{0.7574 \text{ kg H}_2\text{O}}$$
$$= 7.701 \text{ } m$$

(d) the molarity of H₂SO₄ in this solution.

$$M = \frac{\text{moles solute}}{\text{L solution}} = \frac{5.833 \text{ mol}}{1 \text{ L}}$$
$$= 5.833 \text{ } M$$



A solution contains Cu^{2+} (63.55 g/mol) ions at a concentration of 3.0×10^{-4} M. What is the Cu^{2+} concentration in ppm?

$$\frac{\text{mol}}{\text{L}} \longrightarrow \frac{\text{mg}}{\text{L}}$$

Answer = 19 ppm



13.6 Colligative Properties

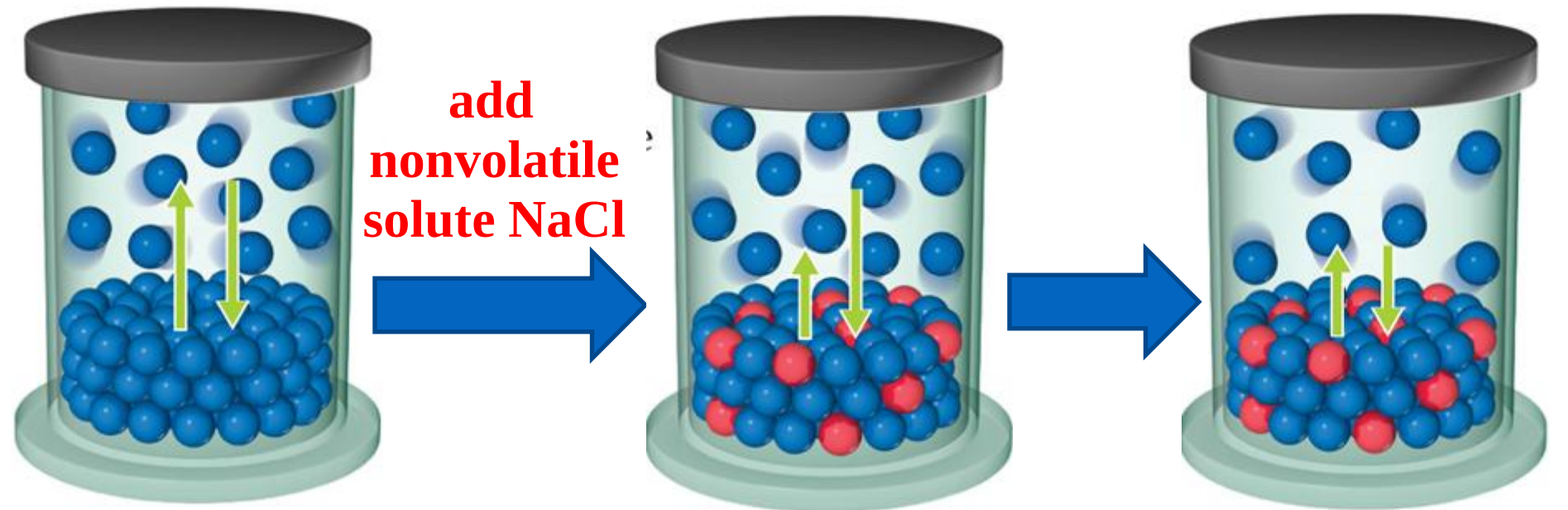
Properties that depend on the **number** (**concentration**) of **solute** **particles** presents in solution, not on the **identity** or **nature** of the solute particles.

Among colligative properties are:

- 1) Vapor-pressure lowering (ΔP)
- 2) Boiling-point elevation (ΔT_b)
- 3) Melting-point depression (ΔT_F)
- 4) Osmotic pressure (π)



Vapor Pressure Lowering



Equilibrium

rate of vaporization
reduced by presence
of NaCl solute

equilibrium reestab-
lished with fewer
H₂O in the gas phase

P°_{solv}

$>$

P_{soln}

Vapor Pressure
of Pure Solvent

Vapor Pressure of
Solvent over Solution



- Because of solute–solvent intermolecular attraction, higher concentrations of nonvolatile solutes make it harder for solvent to escape to the vapor phase.
- Therefore, the vapor pressure of a solution is lower than that of the pure solvent.

Raoult's Law

$$P_{soln} = \chi_{solv} P^{\circ}_{solv}$$

where: χ is the mole fraction of solvent

$$\chi_{solv} = \frac{n_{solv}}{n_{solv} + n_{soln}}$$



If solution contains only one solute:

→ **vapor pressure lowering**

$$\Delta P = P^{\circ}_{solv} - P_{soln} = \chi_{solu} P^{\circ}_{solv}$$

→ **mole fraction of solute**

$$\chi_{solu} = 1 - \chi_{solv}$$

As amount of solute in solution



The mole fraction of solvent



The vapor pressure of solvent over solution



The vapor pressure lowering



What is the vapor pressure of water at 35 °C over a solution made by mixing of

A) 30 g urea with 180 g water at 35 °C?

B) 60 g urea with 180 g water at 35 °C?

C) 90 g urea with 189 g water at 35 °C?

(at 35 °C the vapor pressure of pure water is 42.18 torr, $\mathcal{M}_{\text{urea}} = 60 \text{ g/mol}$, $\mathcal{M}_{\text{H}_2\text{O}} = 18 \text{ g/mol}$)

$$\chi_{\text{solv}} = \frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + n_{\text{urea}}} = \frac{180/18}{(180/18) + (30/60)}$$
$$= \frac{10}{10 + 0.5} \longrightarrow \chi_{\text{solv}} = 0.952$$



$$P_{soln} = \chi_{solv} P^{\circ}_{solv} \longrightarrow P_{soln} = 0.952 \times 42.18$$

$$P_{soln} = 40.17 \text{ torr}$$

$$\Delta P = P^{\circ}_{solv} - P_{soln}$$

$$\Delta P = 42.18 - 40.17 \longrightarrow \Delta P = 2.01 \text{ torr}$$

<i>mass</i>	<i>P</i> [°] _{H₂O}	<i>χ</i> _{solv}	<i>P</i> _{soln}	<i>ΔP</i>
30 g	42.18	0.952	40.17	2.01
60 g	42.18	0.909	38.34	3.84
90 g	42.18	0.870	36.68	5.50



Which of the following aqueous solution of will have the highest vapor pressure lowering?

A) 0.08 m $\text{Al}(\text{NO}_3)_3$

B) 0.1 m NaCl

C) 0.04 m Na_3PO_4

D) 0.08 m Na_2SO_4

E) 0.1 m sucrose

Strong electrolyte

Weak electrolyte



Conc.	Particles / molecule (i)	Highest ΔP	
		<i>m</i>	<i>i \times m</i>
0.08 m $\text{Al}(\text{NO}_3)_3$	4	0.08	0.32
0.1 m NaCl	2	0.10	0.20
0.04 m Na_3PO_4	4	0.04	0.16
0.08 m Na_2SO_4	3	0.08	0.24
0.1 m sucrose	1	0.10	0.10

Lowest ΔP



Which of the following aqueous solution of will have the highest vapor pressure lowering?

A) 0.08 m $\text{Al}(\text{NO}_3)_3$

B) 0.1 m NaCl

C) 0.04 m Na_3PO_4

D) 0.08 m Na_2SO_4

E) 0.1 m sucrose

Strong electrolyte

Weak electrolyte

$\text{Al}(\text{NO}_3)_3 > \text{Na}_2\text{SO}_4 > \text{NaCl} > \text{Na}_3\text{PO}_4 > \text{sucrose}$

highest ΔP

lowest P

lowest ΔP

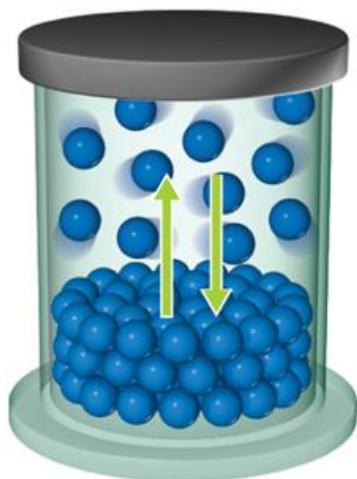
highest P



Boiling Point Elevation

A liquid boils when its vapor pressure equals the atmospheric pressure

solvent

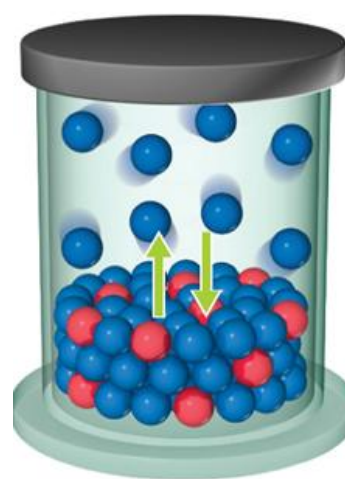


$T_{\text{H}_2\text{O}}$ 100 °C

P_{vapor} 1 atm

will boil

solution



100 °C

< 1 atm

will not boil

to boil increase $T > 100$ °C

Boiling Point Elevation

(K, °C)

$$T_{\text{soln}} - T_{\text{solv}} =$$

$$\Delta T_b = K_b \cdot m$$

molality
mol/kg

molal boiling point
elevation constant

K/m , $K \cdot \text{kg}/\text{mol}$
 $^{\circ}\text{C}/m$, $^{\circ}\text{C} \cdot \text{kg}/\text{mol}$

solvent dependent

Solvent

K_b

Water

0.52

Acetic Acid

3.70

Benzene

2.53

Phenol

3.56

CCl_4

4.02



$$T_{\text{soln}} - T_{\text{solv}} = \Delta T_b = K_b \cdot m$$

As amount of solute in solution 

The molality of solution 

The boiling point elevation 

The boiling point of the solution 



Freezing Point Depression

(K, °C)

$$T_{\text{solv}} - T_{\text{solu}} = \Delta T_F = K_F \cdot m$$

molality
mol/kg

molal freezing point
depression constant

K/m , $K \cdot \text{kg/mol}$
 $^{\circ}\text{C}/m$, $^{\circ}\text{C} \cdot \text{kg/mol}$

solvent dependent

Solvent	K_f
Water	1.86
Acetic Acid	3.90
Benzene	5.12
Phenol	7.27
CCl_4	29.8



$$T_{\text{solv}} - T_{\text{solu}} = \Delta T_F = K_F \cdot m$$

As amount of solute in solution



The molality of solution



The freezing point depression



The freezing of the solution



What is the freezing point of a solution containing 478 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$, antifreeze, 62 g/mol) in 3202 g of water?

$$K_f = 1.86\text{ }^\circ\text{C/m}$$

$$\Delta T_f = K_f \cdot m \quad \longrightarrow \quad \Delta T_f = 1.86\text{ K/m} \times 2.41$$
$$\Delta T_f = 4.48\text{ K}$$

$$\Delta T_f = T_f^\circ - T_f \quad \longrightarrow \quad 4.48\text{ K} = 273 - T_f$$
$$-4.48\text{ }^\circ\text{C} = 268.52\text{ K} = T_f$$

$$n = \frac{478\text{ g}}{62.01\text{ g/mol}} = 7.71\text{ mol}$$

$$m = \frac{\text{mol C}_2\text{H}_6\text{O}_2}{\text{Kg H}_2\text{O}} = \frac{7.71\text{ mol}}{3.202\text{ kg}} = 2.41\text{ m}$$



What is the **molecular weight** of adrenaline if 0.64 grams of adrenaline dissolved in 36.0 g of CCl_4 raises the boiling point by 0.49 degrees Celsius?



Osmotic Pressure:

The minimum pressure that stops osmosis.

$$\pi = MRT$$

π = osmotic pressure in atm

M = molarity of solution

R = gas constant

T = temperature in K

$$\pi_1 = \pi_2$$

isotonic solutions

$$\pi_1 > \pi_2$$

soln₁: hypertonic

concentrated

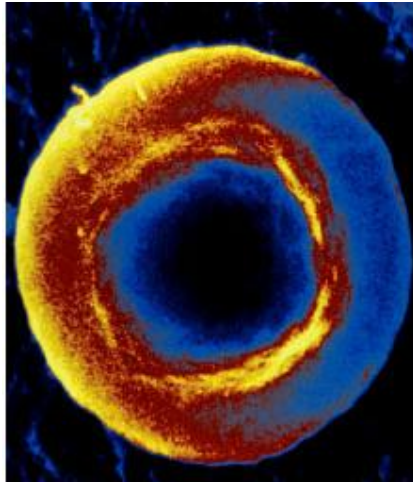
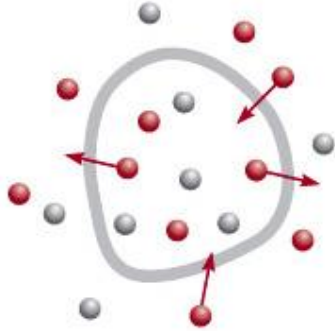
soln₂: hypotonic

diluted

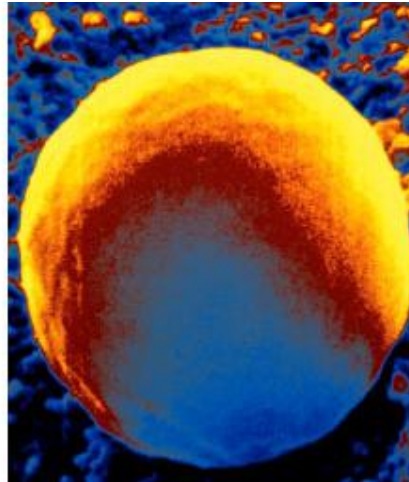
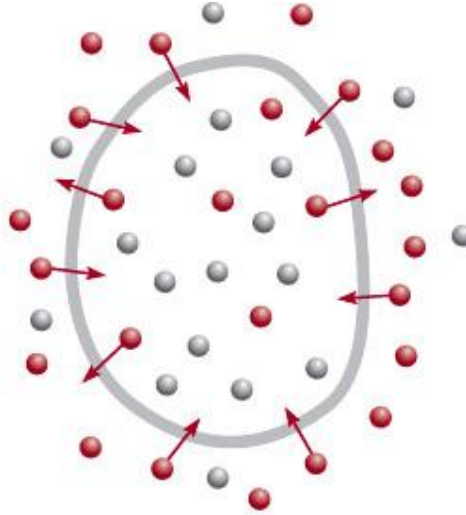


A cell in an:

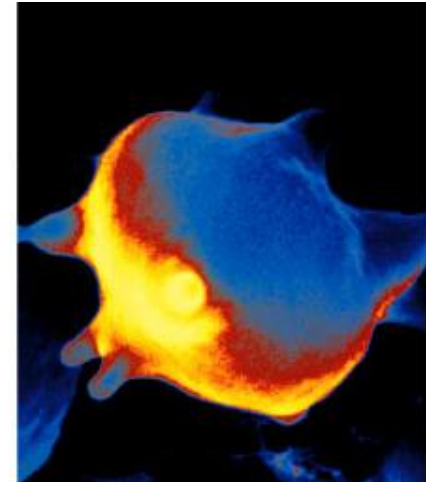
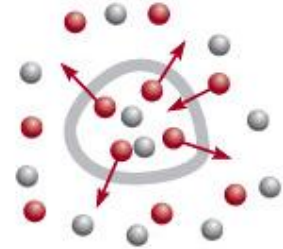
● Water molecules
● Solute molecules



isotonic
solution



hypotonic
solution



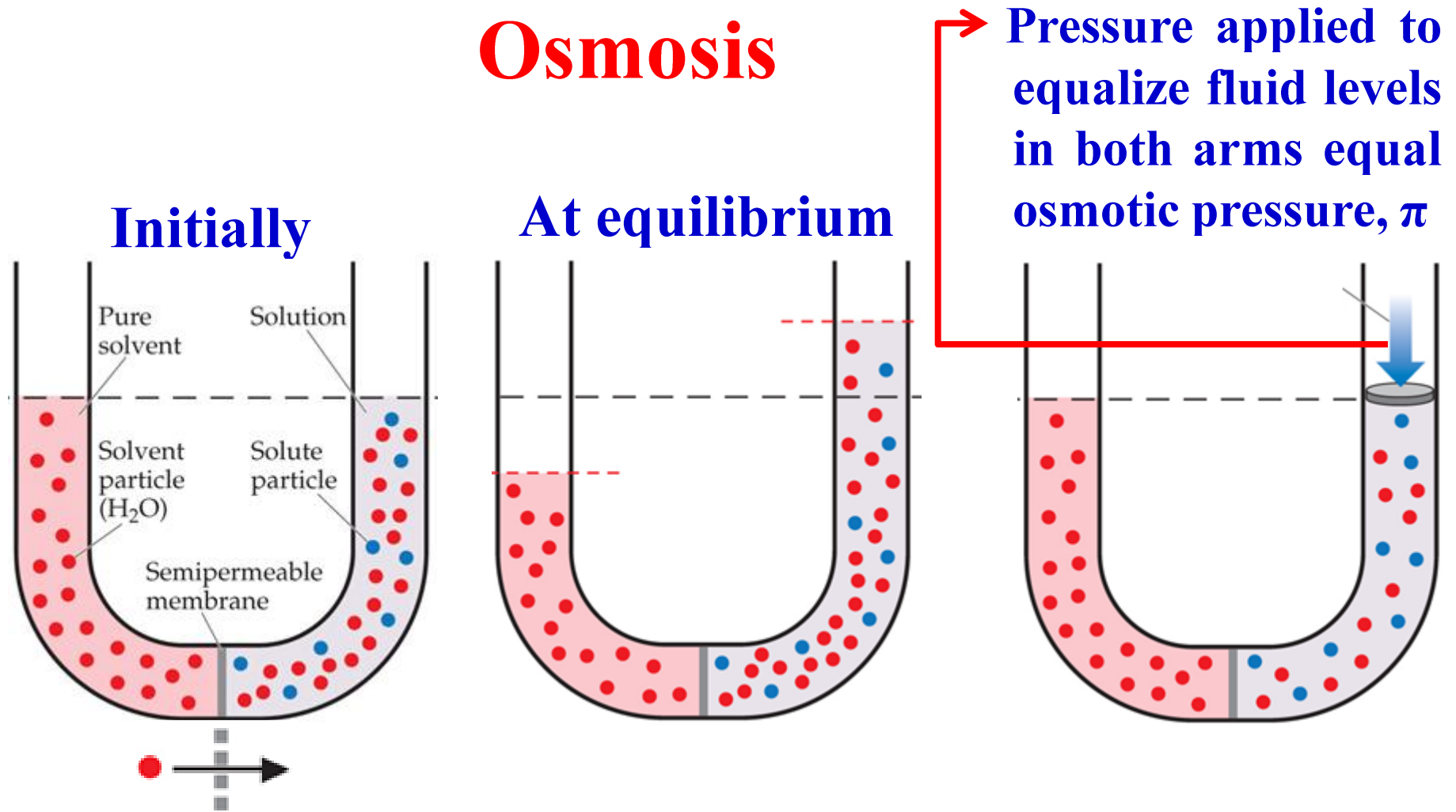
hypertonic
solution

Determination of Molar Mass

Any of the colligative properties can be used for molar mass determination of the solute with osmotic pressure being the most accurate.



Osmosis



In **osmosis**, there is net movement of solvent from the area of **higher solvent concentration** (*lower solute concentration*) to the area of **lower solvent concentration** (*higher solute concentration*).



Osmotic Pressure (π)

The pressure required to stop osmosis, known as **osmotic pressure, π** , is

$$\pi = \frac{\text{mass}}{\mu * V} RT$$

Molar mass of the solute g/mol ← μ

volume of solution (L) ← V

Ideal gas constant ← R

Temp, K ← T

mole solute ← n

$$\pi V = nRT$$

molarity of the solution ← M

$\pi = MRT$



At 298 K, 25.00 mL of solution containing 27.55 mg of protein has an osmotic pressure of 3.22 Torr. The **molecular weight** of the protein is -----.



Which of the following aqueous solution of will have the highest **osmotic pressure**?

A) 1.0 m $\text{Ca}(\text{NO}_3)_2$

B) 3.0 m KCl

C) 2.5 m MgS

D) 4.0 m $\text{C}_6\text{H}_{12}\text{O}_6$

E) H_2O

Strong electrolytes

Weak electrolyte



Conc.	Particles / molecule (i)	<i>m</i>	Highest π <i>i x m</i>
1.0 m $\text{Ca}(\text{NO}_3)_2$	3	1.0	3.0
3.0 m KCl	2	3.0	6.0
2.5 m MgS	2.0	2.5	5.0
4.0 m $\text{C}_6\text{H}_{12}\text{O}_6$	1	4.0	4.0
H_2O	0	0.0	0.0
Lowest π			



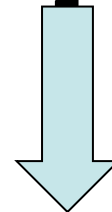
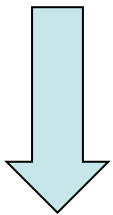
KCl

MgS

Ca(NO₃)₂

C₆H₁₂O₆

H₂O



Highest π
Highest ΔT_b
Highest ΔT_F
Highest ΔP

**Colligative
properties**

Lowest π
Lowest ΔT_b
Lowest ΔT_F
Lowest ΔP

Highest T_b
Lowest T_F
Lowest P

Lowest T_b
Highest T_F
Highest P

